

Capture threshold for electron interacting with circularly polarized waves in parallel static electric fields

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This study investigates the threshold for resonant capture of electrons interacting with circularly polarized electromagnetic waves in a static electric field parallel to the background magnetic field. Captured electrons can channel energy supplied by the static electric field into perpendicular cyclotron motion while maintaining resonance. An analytical expression for the critical wave amplitude required for capture is derived using a nonlinear Schrödinger-type equation, characterizing the transition between passage through resonance and sustained resonant capture. The analytical prediction is compared directly with numerical results, showing strongly agreement. Capture conditions for right- and left-hand circularly polarized waves are also analyzed to examine the influence of wave polarization on resonant electron dynamics.

Key words: Autoresonance, Capture Threshold, Anomalous Doppler resonance, Normal Doppler resonance

1. Introduction

Resonant interactions between electrons and electromagnetic waves govern energy exchange and pitch-angle evolution in magnetized plasmas. Coherent whistler waves can produce substantial, nondiffusive changes in energetic-electron trajectories, with implications for magnetospheric particle dynamics (Yoon & Bellan 2020). In tokamak plasmas, direct observations of runaway-electron-driven whistler waves have established a connection between electromagnetic fluctuations and energetic-electron populations (Spong *et al.* 2018). Externally injected whistler waves have also been proposed as a means of controlling runaway-electron energy through resonant scattering (Guo *et al.* 2018). More recently, experiments in DIII-D demonstrated pitch-angle scattering and suppression of the high-energy electron population following the application of externally launched helicon waves (Choudhury *et al.* 2026). These developments motivate a quantitative understanding of the conditions under which coherent waves can sustain resonant interactions with electrons subject to electric-field acceleration.

In a coherent wave-particle interaction, the relative phase between the wave and electron gyromotion determines whether an electron passes through resonance or remains phase locked. Wave polarization and the Doppler-shifted frequency are therefore central to the accessible resonance and the resulting energy exchange. For propagation parallel to the background magnetic field, the fundamental normal and anomalous Doppler resonances are distinguished by the sign of the Doppler-shifted wave frequency. In the

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anomalous resonance, parallel motion can supply energy to perpendicular gyromotion and the wave. Xu *et al.* (2025) examined this process through angular-momentum conservation and established a correspondence between quantum descriptions and classical simulations of energy transfer. A parallel static electric field adds a further source of energy and changes the electron's perpendicular cyclotron motion, making the onset and persistence of resonant capture sensitive to both the wave polarization and the imposed acceleration.

The underlying single-particle dynamics have a long theoretical history. Roberts & Buchsbaum (1964) obtained exact relativistic solutions for a charged particle interacting with a transverse electromagnetic wave propagating along a uniform magnetic field, including the effects of wave-induced parallel motion and the relativistic variation of the cyclotron frequency. Schram & Beukema (1969) explicitly investigated an additional parallel electrostatic field and identified a transition between sustained average energy growth and passage through resonance, with a transition condition depending on the wave amplitude and refractive index. Related work on gyroresonant surfing showed that an electrostatic force can sustain cyclotron resonance with a monochromatic circularly polarized wave and enable substantial particle energization (Kuramitsu & Krasnoselskikh 2005). These results establish that a parallel electric field can modify the duration and character of resonant energy exchange.

Nonlinear formulations provide further insight into the trajectories associated with strong wave-particle coupling. Bellan (2013) expressed the relativistic dynamics in terms of a pseudo-potential depending on the frequency mismatch, providing a framework for interpreting large pitch-angle changes. Yoon & Bellan (2020) subsequently used this approach to describe nondiffusive scattering of particle distributions by coherent whistler waves. For a right-hand circularly polarized wave in the presence of a constant parallel electric field, Kang *et al.* (2026) demonstrated a reversal of parallel acceleration accompanied by perpendicular energization after resonant trapping. Their study left the exact trapping condition for future investigation. This motivates an explicit analysis of the transition into capture, complementing descriptions of the subsequent trapped motion.

Autoresonance provides a natural framework for examining this transition. In an autoresonant state, nonlinear frequency adjustment allows an oscillator to remain phase locked while the resonance parameters evolve slowly. The principle of sustained synchronism has historical roots in accelerator phase stability (McMillan 1945). Its relevance to electron cyclotron heating was examined by Neishtadt & Timofeev (1987), who studied capture and acceleration in inhomogeneous magnetic traps. Their analysis illustrates that autoresonance can occur at a fixed laboratory wave frequency when particle motion changes the local resonance condition. A broader account of autoresonant excitation and its applications to nonlinear oscillators was presented by Fajans & Friedland (2001).

A central result of autoresonance theory is the existence of a finite drive threshold for capture during resonance passage. For a broad class of weakly nonlinear oscillators with a linear frequency sweep and sufficiently small initial amplitude, the critical drive scales as the three-quarter power of the sweep rate (Fajans & Friedland 2001). Friedland (2008) formulated the capture dynamics using a driven nonlinear Schrödinger-type equation and examined how the threshold depends on the initial point of the resonance passage. Further studies quantified the influence of damping (Fajans *et al.* 2001) and showed how thermal fluctuations broaden the capture transition while self-fields modify its width (Barth *et al.* 2009). The conversion of a normalized threshold into a physical wave-amplitude criterion consequently requires explicit specification of the reduction, normalization and initial conditions.

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 2. This is the second item of the inner list. LATEX allows you to nest lists deeper than you really should.

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The default formatting of numbering can be changed as (1), (2), (3). . . , for example,

- (1) List entry one.
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- (a) List entry.
- (b) List entry two.
- (c) List entry three.
- (d) List entry four.
- (e) List entry five.

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TABLE 1. Table caption

Year	Single outlet	Small multiple ^a	Large multiple	Total
1982 ^b	98	129	620	847
1987	138	176	1000	1314
1991	173	248	1230	1651
1998	200	300	1500	2000

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^bThis is an example of second numbered tablenote

1.4. Equations

Example of a single-line equation 1.1.

$$a = b \text{ ((Single Equation Numbered))} \dots \quad (1.1)$$

Equations can also be multiple lines as shown in Equations 1.2 and 1.3.

$$c = 0 \text{ ((Multiple Lines, Numbered))} \dots \quad (1.2)$$

$$ac = 0 \text{ ((Multiple Lines, Numbered))} \dots \quad (1.3)$$

Unnumbered display equations single line:

$$\Gamma\psi' = x'' + y^2 + z_i^n$$

Unnumbered display equations multiple lines:

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$ac = 0$ ((Multiple Lines, Unnumbered))...

1.4.1. *Third-Order Heading*

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TABLE 2. Table caption table caption table caption table caption table caption table
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460	2.12	1.04	8.63	3.48
476	2.24	0.94	9.22	3.73
676	3.71	1.67	16.36	6.00
572	2.86	1.29	11.90	4.56

^aFirst Table Footnote;^bTSecond Table Footnote.

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THEOREM 1. *This is an example of enunciation. This is an example of enunciation. This is an example of enunciation. This is an example of enunciation. This is an example of enunciation. This is an example of enunciation.*

$$AB = \sum_i a_i b_i$$

2. FIRST-ORDER HEADING

Fourth-Order Heading

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$$AB = \sum_i a_i b_i$$

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THEOREM 3 (MATH DETAIL). *This is an example of enunciation. This is an example of enunciation. This is an example of enunciation. This is an example of enunciation. This is an example of enunciation.*

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Acknowledgements

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Appendix

First order heading

TABLE 3. Table caption table caption table caption table caption table caption table caption table caption table caption table caption table caption table caption table caption table caption

Column One	Column Two	Span Col 3 & 4		Span Col 5 & 6	
		Column Three	Column Four	Column Five	Column Six
6-31+G(d,p)	460	2.12	1.04	8.63	3.48
6-31++G(d,p)	476	2.24	0.94	9.22	3.73
6-31++G(df,p)	676	3.71	1.67	16.36	6.00
6-311+G(d,p)	572	2.86	1.29	11.90	4.56
6-311++G(3df,2p)	1,060	7.54	3.25	36.70	11.45

^{a)}Table Footnote

^{b)}Second Table Footnote.
